

Task 4: Restore an EC2 Instance from the Latest Snapshot

Step 1: Create an IAM Role

Create an IAM role with the following configuration:

- **Trusted Entity:** AWS Service
- **Use Case:** Lambda
- **Permissions Policy:** **AWSLambdaBasicExecutionRole**
- **Role Name:** **S3BucketCleanupRole**
- Add an inline policy to the **S3BucketCleanupRole** IAM role with the following JSON policy.**Policy Name:** **Ec2snapshot**

The screenshot displays the AWS IAM console for the **S3BucketCleanupRole**. At the top, a green notification bar states "Policy Ec2snapshot created." The role's description is "Allows Lambda functions to call AWS services on your behalf." The **Summary** tab is active, showing the creation date as July 23, 2026, 12:49 (UTC+05:30), and the last activity as 4 hours ago. The ARN is `arn:aws:iam::279473426275:role/S3BucketCleanupRole` and the maximum session duration is 1 hour. Below the summary, the **Permissions** tab is selected, showing two attached policies: **AWSLambdaBasicExecutionRole** (AWS managed, 3 attached entities) and **Ec2snapshot** (Customer inline, 0 attached entities). A section for "Generate policy based on CloudTrail events" is also visible, with a "Generate policy" button and a note that no requests have been made in the past 7 days.

Step 2: Create Lambda Function

Create an AWS Lambda function with the following configuration:

Function Name: **EC2SnapRestore**

Runtime: Python 3.14

Execution Role: **S3BucketCleanupRole**

EC2SnapRestore

Throttle Copy ARN Actions

Function overview Info

Diagram Template

EC2SnapRestore

Layers (0)

+ Add trigger + Add destination

Function ARN: arn:aws:lambda:us-east-1:279473426275:function:EC2SnapRestore

Description: -

Last modified: 21 seconds ago

Expert to Infrastructure Composer Download

Code Test Monitor Configuration Aliases Versions

Configuration

General configuration Triggers Permissions Destinations Function URL Environment variables Tags VPC RDS databases Monitoring and operations tools Concurrency and recursion detection Asynchronous invocation Code signing File systems

Execution role

Role name: S3BucketCleanupRole

Resource summary

To view the resources and actions that your function has permission to access, choose a service.

Amazon CloudWatch Logs (3 actions, 1 resource)

By action By resource

Resource Actions

All resources

Allow: logs:CreateLogGroup
Allow: logs:CreateLogStream
Allow: logs:PutLogEvents

Lambda obtained this information from the following policy statements:

- Managed policy AWSLambdaBasicExecutionRole, statement 0

Resource-based policy statements

View policy Edit Delete Add permissions

Step 3: Create a Test Event

Create a test event to invoke the Lambda function with the following configuration:

- Event Name: EC2SnapLog
- Event JSON:

```
{
```

The test event "EC2SnapLog" was successfully saved.

Function overview

Diagram

Template

EC2SnapRestore

Layers (0)

+ Add trigger

+ Add destination

Function ARN

arn:aws:lambda:us-east-1:279473426275:function:EC2SnapRestore

Description

-

Last modified

4 minutes ago

Export to Infrastructure Composer

Download

Code

Test

Monitor

Configuration

Aliases

Versions

Executing function: succeeded

Details

Test event

Delete

CloudWatch Logs Live Tail

Save

Test

To invoke your function without saving an event, modify the event, then choose Test. Lambda uses the modified event to invoke your function, but does not overwrite the original event until you choose Save.

Test event action

Create new event

Edit saved event

Invocation type

Synchronous

Asynchronous

Event name

EC2SnapLog

Event JSON

Format JSON

Step 4 — Find Required Information

An EC2 instance has an Instance ID. Go to the Storage tab to get the Volume ID, and then go to the Networking tab to find the Security Group ID and Subnet ID.

☒	travelmemory-server-1	i-06039e8206d49f6a9	Running	t2.micro	2/2 checks passed	us-east-1c	ec2-54-172-156-49.co...	54.172.156.49	-	-	disabled
☐	projecttest1	i-0d3123c2eb21127a9	Stopped	t2.micro	-	us-east-1c	-	-	-	-	disabled
☐	travelmemory-server-2	i-0800c2b13bea55a35	Stopped	t2.micro	-	us-east-1c	-	-	-	-	disabled
☐	Restored-EC2	i-01e9e23ee65d6d156	Running	t3.micro	Initializing	us-east-1c	ec2-100-31-208-78.co...	100.31.208.78	-	-	disabled
☐	groupSec2	i-0ecfa209aace842f5	Stopped	t2.micro	-	us-east-1b	-	-	-	-	disabled

i-06039e8206d49f6a9 (travelmemory-server-1)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Root device details

Root device name

/dev/sda1

Root device type

EBS

EBS optimization

disabled

Block devices

Filter block devices

Volume ID	Device name	Volume size (GiB)	Volume State	Attachment status	EBS card index	Attachment time	Encrypted	KMS key ID	Delete on termination	
☐	vol-0a764c6e9f5c2981f	/dev/sda1	20	In-use	Attached	0	2026/07/01 16:16 GMT+5:30	No	-	Yes

Volume monitoring (1)

Investigate with AI - new

1h

3h

12h

1d

3d

1w

Custom

UTC timezone

Explore related

No widget on this dashboard.

Documentation

☒	travmemory-server-1	i-06039e8206d49f6a9	Running			t2.micro	2/2 checks pass	us-east-1c	ec2-54-172-156-49.co...	54.172.156.49	-	-	disabled
☐	projecttest1	i-0d3123c2eb21127a9	Stopped			t2.micro	-	us-east-1c	-	-	-	-	disabled
☐	travmemory-server-2	i-0800c2b13bea55a35	Stopped			t2.micro	-	us-east-1c	-	-	-	-	disabled
☐	Restored-EC2	i-01e9e23ee65d6d156	Running			t3.micro	Initializing	us-east-1c	ec2-100-31-208-78.co...	100.31.208.78	-	-	disabled
☐	groupSec2	i-0ecfa209aace842f3	Stopped			t2.micro	-	us-east-1b	-	-	-	-	disabled

i-06039e8206d49f6a9 (travmemory-server-1)

Details | Status and alarms | Monitoring | Security | **Networking** | Storage | Tags

VPC ID
vpc-0b556f972f55248a6

Availability zone ID
use1-az4

Public IP addresses
Public IPv4 address
54.172.156.49 | open address

Secondary private IPv4 addresses
-

▼ Hostname and DNS
Public DNS
ec2-54-172-156-49.compute-1.amazonaws.com | open address
Dualstack - IP based name: A and AAAA record
Private hostname type

Subnet ID
subnet-0e002f0a3f6f9112

Outpost ID
-

Private IP addresses
Private IPv4 addresses
172.31.29.172
Carrier IP addresses (ephemeral)
-

Private IP DNS name (IPv4 only)
ip-172-31-29-172.ec2.internal
IPv6-only - IP based name: AAAA record only
-
Use RBN as guest OS hostname

Availability zone
us-east-1c

IPv6 addresses
-

IPv4-only IP based name: A record only
ec2-54-172-156-49.compute-1.amazonaws.com
Public hostname type
public-ipv4-dns-name
Answer RBN DNS hostname IPv4

☒	travmemory-server-1	i-06039e8206d49f6a9	Running			t2.micro	2/2 checks pass	us-east-1c	ec2-54-172-156-49.co...	54.172.156.49	-	-	disabled
☐	projecttest1	i-0d3123c2eb21127a9	Stopped			t2.micro	-	us-east-1c	-	-	-	-	disabled
☐	travmemory-server-2	i-0800c2b13bea55a35	Stopped			t2.micro	-	us-east-1c	-	-	-	-	disabled
☐	Restored-EC2	i-01e9e23ee65d6d156	Running			t3.micro	Initializing	us-east-1c	ec2-100-31-208-78.co...	100.31.208.78	-	-	disabled
☐	groupSec2	i-0ecfa209aace842f3	Stopped			t2.micro	-	us-east-1b	-	-	-	-	disabled

i-06039e8206d49f6a9 (travmemory-server-1)

Details | Status and alarms | Monitoring | **Security** | Networking | Storage | Tags

▼ Security details
IAM role
-
Security groups
sg-09a958cf2d9acdfaf (travmemory-sg)

Owner ID
279473426275

Launch time
Sat Jul 25 2026 11:18:56 GMT+0530 (India Standard Time)

▼ Inbound rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sgr-0a62ff8569885f113	80	TCP	0.0.0.0/0	travmemory-sg	Web traffic
-	sgr-0594f1f51af32eb5c	22	TCP	0.0.0.0/0	travmemory-sg	Secure SSH access
-	sgr-01debfa319f4900b8	3000	TCP	0.0.0.0/0	travmemory-sg	WEB 1
-	sgr-0c866ec3055f19dd2	443	TCP	0.0.0.0/0	travmemory-sg	Secure web traffic

Step 5 — Testing Event

Execute the Lamda function and Click on the Test

Successfully updated the function "EC2SnapRestore".

+ Add trigger

+ Add destination

13 hours ago

Code

Test

Monitor

Configuration

Aliases

Versions

Executing function: succeeded (logs)

Details

```
{
  "statusCode": 200,
  "Snapshot": "snap-0bb0ed7877447178b",
  "AMI": "ami-0ece1b9d50ef57482",
  "Instance": "i-01e9e23ee65d6d156"
}
```

Summary

Code SHA-256

sKznVCVUT13KvjTFeQk7y8zkPbTuuOkstf0do+3g=

Function version

\$LATEST

Duration

2290.31 ms

Resources configured

128 MB

Init duration

550.40 ms

Execution time

14 seconds ago

Request ID

4bf4ae08-5fef-485b-bd17-07ded6993220

Billed duration

2841 ms

Max memory used

100 MB

Log output

The area below shows the last 4 KB of the execution log. [Click here](#) to view the corresponding CloudWatch log group.

```
START RequestId: 4bf4ae08-5fef-485b-bd17-07ded6993220 Version: $LATEST
AMI Created: ami-0ece1b9d50ef57482
New Instance: i-01e9e23ee65d6d156
END RequestId: 4bf4ae08-5fef-485b-bd17-07ded6993220
REPORT RequestId: 4bf4ae08-5fef-485b-bd17-07ded6993220 Duration: 2290.31 ms Billed Duration: 2841 ms Memory Size: 128 MB Max Memory Used: 100 MB Init Duration: 550.40 ms
```

Test event info

Delete

CloudWatch Logs Live Tail

Save

Test

Step 6: Verify Instance

On the EC2 Instances we can see Restored EC2 instance

Instances (1/6)

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring
travellmemory-server-second	i-0600dd280af17esa	Stopped	t2.micro		us-east-1c	-	-	-	-	disabled
travellmemory-server-1	i-06039e8206d49fa9	Running	t2.micro	2/2 checks passed	us-east-1c	ec2-54-172-156-49.co...	54.172.156.49	-	-	disabled
projecttest1	i-0d3123c2eb21127a9	Stopped	t2.micro		us-east-1c	-	-	-	-	disabled
travellmemory-server-2	i-0800c2b13bea55a35	Stopped	t2.micro		us-east-1c	-	-	-	-	disabled
Restored-EC2	i-01e9e23ee65d6d156	Running	t3.micro	Initializing	us-east-1c	ec2-100-31-208-78.co...	100.31.208.78	-	-	disabled
groupSec2	i-0ecfa209aace842f3	Stopped	t2.micro		us-east-1b	-	-	-	-	disabled

i-01e9e23ee65d6d156 (Restored-EC2)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID

i-01e9e23ee65d6d156

IPv6 address

-

Hostname type

IP name: ip-172-31-25-210.ec2.internal

Answer private resource DNS name

-

Auto-assigned IP address

100.31.208.78 [Public IP]

IAM role

-

IMDSv2

Optional

Public IPv4 address

100.31.208.78 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-25-210.ec2.internal

Instance type

t3.micro

VPC ID

vpc-0b556f972f55248a6

Subnet ID

subnet-0e02f0a3ff6f9112

Instance ARN

arn:aws:ec2:us-east-1:279473426275:instance/i-01e9e23ee65d6d156

Private IPv4 addresses

172.31.25.210

Public DNS

ec2-100-31-208-78.compute-1.amazonaws.com | open address

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

Step 7: Validate

Checked all the configuration and the volumes, Subnet ID,

Security Group ID
Availability Zone

☒	Restored-EC2	i-01e9e23ee65d6d156	Running	t3.micro	Initializing	us-east-1c	ec2-100-31-208-78.co...	100.31.208.78	-	-	disabled
☐	groupSec2	i-0ecfa209aace842f3	Stopped	t2.micro	-	us-east-1b	-	-	-	-	disabled

i-01e9e23ee65d6d156 (Restored-EC2)

DetailsStatus and alarmsMonitoringSecurityNetworkingStorageTags

▼ IPC ID
vpc-0b556f972f55248a6

▼ Availability zone ID
use1-az4

▼ IP addresses
Public IPv4 address
100.31.208.78 | open address
Secondary private IPv4 addresses

▼ Hostname and DNS
Public DNS
ec2-100-31-208-78.compute-1.amazonaws.com | open address
Dualstack - IP based name: A and AAAA record
Private hostname type

Subnet ID
subnet-0e02f0a3ff6f9112

Outpost ID

Private IPv4 addresses
172.31.25.210
Carrier IP addresses (ephemeral)

Private IP DNS name (IPv4 only)
ip-172-31-25-210.ec2.internal
IPv6-only - IP based name: AAAA record only
Use RBN as guest OS hostname

Availability zone
us-east-1c

IPv6 addresses

IPv4-only IP based name: A record only
ec2-100-31-208-78.compute-1.amazonaws.com
Public hostname type
public-ipv4-dns-name
Answer RBN DNS hostname IPv4

☒	Restored-EC2	i-01e9e23ee65d6d156	Running	t3.micro	Initializing	us-east-1c	ec2-100-31-208-78.co...	100.31.208.78	-	-	disabled
☐	groupSec2	i-0ecfa209aace842f3	Stopped	t2.micro	-	us-east-1b	-	-	-	-	disabled

-01e9e23ee65d6d156 (Restored-EC2)

DetailsStatus and alarmsMonitoringSecurityNetworkingStorageTags

▼ Root device details
Root device name
/dev/xvda
Block devices

Root device type
EBS
EBS optimization
disabled

Filter block devices

Volume ID	Device name	Volume size (GiB)	Volume State	Attachment status	EBS card index	Attachment time	Encrypted	KMS key ID	Delete on termination
☒ vol-0a6db87c16cb7af6f	/dev/xvda	20	In-use	Attached	0	2026/07/25 11:28 GMT+5:30	No	-	Yes

Volume monitoring (1)

Alarm recommendations

Investigate with AI - new

1h3h12h1d3d1wCustom

UTC timezone

Explore related

Stalled I/O Check

Average read latency (ms/op)

Average write latency (ms/op)

Read throughput (KiB/s)

☒	Restored-EC2	i-01e9e23ee65d6d156	Running	t3.micro	Initializing	us-east-1c	ec2-100-31-208-78.co...	100.31.208.78	-	-	disabled
☐	groupSec2	i-0ecfa209aace842f3	Stopped	t2.micro	-	us-east-1b	-	-	-	-	disabled

-01e9e23ee65d6d156 (Restored-EC2)

DetailsStatus and alarmsMonitoringSecurityNetworkingStorageTags

▼ Security details
IAM role
ec2-instance-profile-role
Security groups
sg-09a9558cf2d9acd4f (travelmemory-sg)

Owner ID
79473426275
Launch time
Sat Jul 25 2026 11:28:54 GMT+0530 (India Standard Time)

▼ Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sgr-0a62ff8569885f113	80	TCP	0.0.0.0/0	travelmemory-sg	Web traffic
-	sgr-0594f1f51af32eb5c	22	TCP	0.0.0.0/0	travelmemory-sg	Secure SSH access
-	sgr-01debfa319f4900b8	3000	TCP	0.0.0.0/0	travelmemory-sg	WEB 1
-	sgr-0c866ec305f19dd2	443	TCP	0.0.0.0/0	travelmemory-sg	Secure web traffic